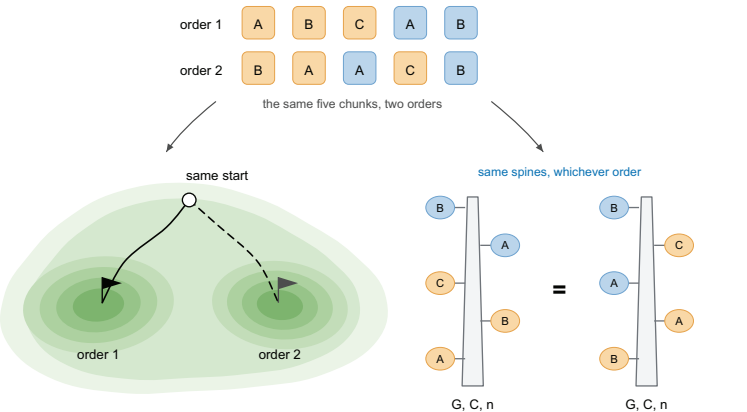
a **Order cannot interfere: a sum does not remember the order**



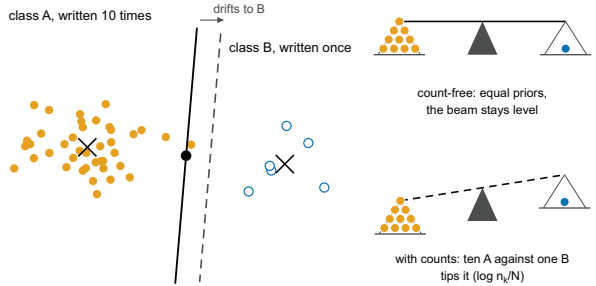
loss landscape of a gradient-based learner

the weights follow the path: two orders, two minima; the last chunks overwrite the first

write-only memory: a sum over the written chunks

each write adds a spine the dendrite keeps (the synapse of Fig. 1b); a sum is a symmetric function of its terms: the same sums whichever order, so the same readout (Proposition 1)

b **Repetition cannot bias: the readout counts nothing**



crosses: class means; dot: their midpoint
solid: count-free discriminant; equal priors, shared covariance (Prop. 2)
dashed: with class counts, log(n_k/N), the boundary drifts towards the rarer class

$$s_k(h) = h^T P \mu_k - \tfrac{1}{2} \mu_k^T P \mu_k$$

no log n_k term